

Federal Urdu University of Arts Science and Technology

LAB 4

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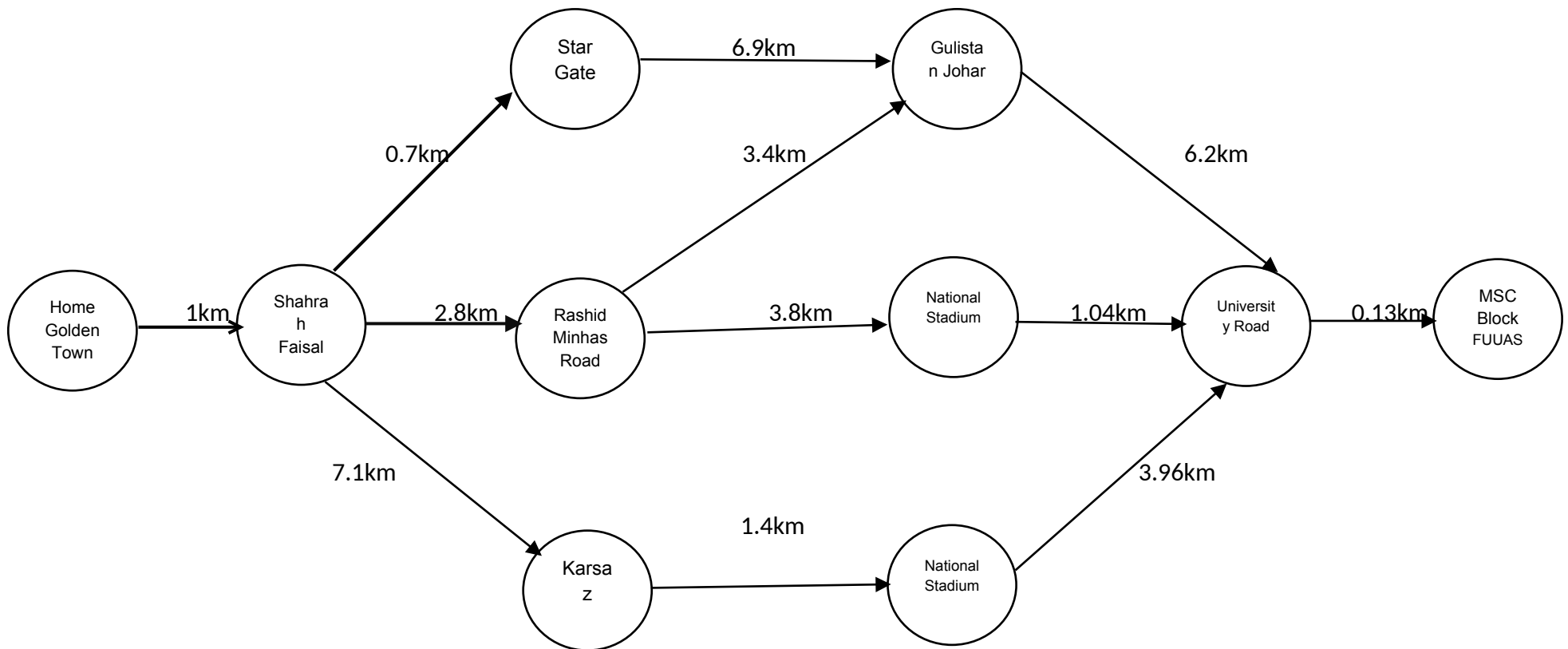
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Semester: VI

Section: BSCS-B

A map is drawn showing the route from home (Golden Town Shah Faisal Colony) to the Federal Urdu University of Arts, Science and Technology (FUUAST), Computer Science Department (MSC Block). The road network is represented as a weighted graph, where each intersection or landmark is a node and each road segment is an edge labelled with its distance in kilometers. Four possible routes are identified between the source (Home) and the destination (FUUAST)

Route Graph:



Possible Routes:

ROUTE 1

Home → Shahrah-e-Faisal → Rashid Minhas → NIPA → University Road → FUUAST

ROUTE 2

Home → Shahrah-e-Faisal → Star Gate/Airport → Gulistan-e-Johar → University Road → FUUAST

ROUTE 3

Home → Shahrah-e-Faisal → Karsaz → National Stadium → University Road → FUUAST

ROUTE 4

Home → Shahrah-e-Faisal → Rashid Minhas → Gulistan-e-Johar → University Road → FUUAST

Edge Weighted Graph:

FROM	TO	DISTANCE IN KM
Golden Town	Shahrah-e-Faisal	1
Shahrah-e-Faisal	Star Gate / Airport	0.7
Shahrah-e-Faisal	Karsaz	7.1
Shahrah-e-Faisal	Rashid Minhas Road	2.8
Star Gate / Airport	Gulistan-e-Johar	6.9
Rashid Minhas Road	Gulistan-e-Johar	3.4
Rashid Minhas Road	NIPA	3.8
Karsaz	National Stadium	1.4
NIPA	University Road	1.04
National Stadium	University Road	3.96
Gulistan-e-Johar	University Road	6.2
University Road	MSC Block, FUUAST	0.13

Route Distance:

ROUTE 1 = 8.77 km

ROUTE 2 = 14.93 km

ROUTE 3 = 13.59 km

ROUTE 4 = 13.53 km

Shortest Route:

Route 1 has the smallest total distance: 8.77 km. Therefore, Route 1 is the shortest route in this weighted graph.